

Where Will the Penalty Go?

Kicker placement and goalkeeper dives in 1,300+ penalties from public data

Abstract. A penalty is an asymmetric guessing game between kicker and goalkeeper. Using 1,032 open-play and shootout penalties (StatsBomb) and 320 shootout kicks with coded dives (Kaggle), we report three results. First, kicker direction is close to unpredictable beyond footedness, so the value of a probabilistic model lies in calibration rather than top-1 accuracy. Second, goalkeepers show a strong static directional bias but no measurable sequential dependence. Third, combining the two implies a modest, largely unexploited scoring edge from shooting to the non-natural side. Each result is reported with explicit uncertainty.

1,032

320

88%

0.78 vs 0.71

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StatsBomb penalties shootout kicks with dives of keepers commit to a side P(score): non-natural vs natural sequential tests null

Data and methods

Data. The kicker model uses StatsBomb open data (1,032 penalties, 534 kickers, 14 competitions; placement L/C/R from the shot end-location). The goalkeeper analysis uses the Kaggle World Cup shootouts, 1982 to 2022 (35 shootouts, 320 kicks), because StatsBomb does not record dive direction: the keeper is logged at the set point on the goal line, which we confirmed on the raw events including saved penalties.

Coding. The two corpora label left and right from opposite viewpoints. Placement and dive are therefore recoded as **natural**, **centre**, or **non-natural** relative to the kicker's foot, which is frame-invariant and the only consistent basis for combining the sources.

Models and tests. Kicker direction is modelled with a Bayesian hierarchical multinomial logit (a footedness baseline, a shootout indicator, and a shooter random intercept for partial pooling), sampled with nutpie (4 chains, 2,000 draws; maximum R-hat 1.005). It is compared against the footedness marginal, a per-player frequency table, and a logit without random effects, and scored on log-loss, Brier score, and expected calibration error (ECE), with an exact-binomial McNemar test, forward-chaining temporal cross-validation, and a sensitivity sweep over the centre-band width. Goalkeeper sequential dependence is tested with an exact binomial test, a permutation test, and bootstrap intervals, with explicit statistical power. The placement-by-dive payoff table and the expected-score comparison use a cluster bootstrap over shootouts, and reported proportions carry Wilson 95% intervals.

1. Kicker direction: footedness dominates, and the model's value is calibration

At the population level, placement is largely determined by footedness: right-footers favour one side and left-footers the mirror image, with an overall split of 50% natural, 14% centre, 36% non-natural. Beyond footedness, direction is close to random, and top-1 accuracy is bounded near 50% because takers mix near the game-theoretic equilibrium. A model therefore cannot reliably out-guess the kicker; its value lies in well-calibrated probabilities, in the sense that predictions assigned 80% probability should occur about 80% of the time.

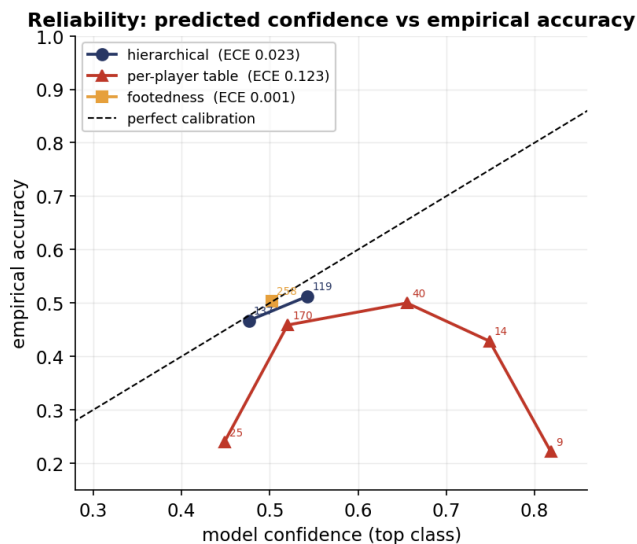


Figure 1: Reliability of each predictor. The per-player frequency table (red) is markedly over-confident on low-sample takers; the hierarchical model and the footedness marginal lie near the diagonal. ECE in parentheses.

The hierarchical model is well calibrated (ECE 0.02) and clearly outperforms the per-player table (ECE 0.12), which is badly over-confident on the many takers with few recorded penalties. It also attains lower out-of-sample log-loss than the table at both sample-size extremes (low- n 1.01 vs 1.11; high- n 0.94 vs 1.02; exact McNemar $p = 0.047$). The mechanism is partial pooling: each player's estimate is drawn toward his own record only in proportion to the evidence behind it. Individual tendencies are real but small: the shooter random-effect scale is non-zero (posterior about 0.4, 94% interval excluding zero), so placement is mostly footedness plus a modest personal lean that is well identified only for high-volume takers. One contextual effect is robust: under shootout pressure the natural-side share rises from 49% to 55%, a shift in the placement distribution that is distinct from the documented effects of pressure on conversion.

2. Goalkeeper behaviour: a strong static bias, no sequential signal

Goalkeepers are far more predictable than kickers, but only in aggregate. They commit to a side on 88% of kicks and lean toward the kicker's natural side. They show no usable sequential structure: across four hypotheses, the gambler's fallacy (diving opposite a streak), dependence on the previous kick, autocorrelation between successive dives, and win-stay/lose-shift on the previous outcome, every effect is null (all $p > 0.4$). The lag-1 test is adequately powered (176 cases; 99% power against a true 65% tendency) and returns 51%, so this is evidence of absence rather than absence of evidence, which addresses the long-standing dispute over the gambler's fallacy in shootouts on modern data.

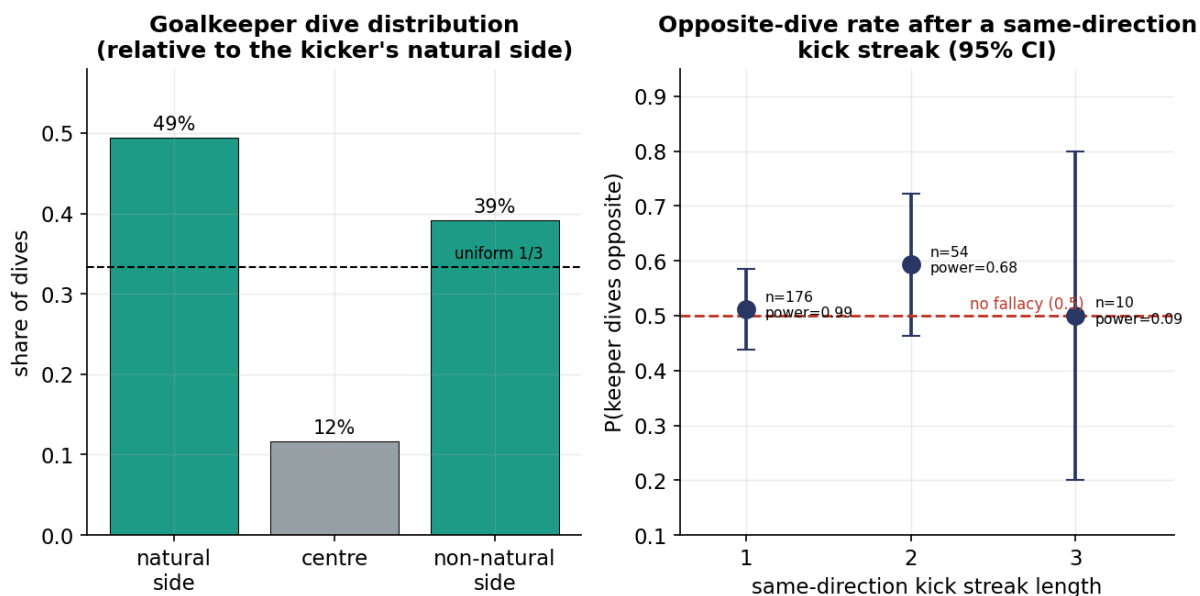


Figure 3: Left: the goalkeeper dive distribution (relative to the kicker's natural side). Right: the rate of diving opposite after a same-direction kick streak, with bootstrap 95% intervals; all intervals include 0.5, and the lag-1 case is well powered.

The exploitable quantity is therefore the static directional bias, not any in-shootout pattern.

3. Exploiting the goalkeeper's directional bias

Because keepers over-commit to the natural side, the placement-by-dive payoff table is asymmetric: the keeper saves chiefly when his dive matches the kick, leaving the non-natural side comparatively under-defended. Expected score is highest for the non-natural corner (0.78, against 0.71 natural and 0.66 centre), and the realized rates agree (0.75, 0.67, 0.64).

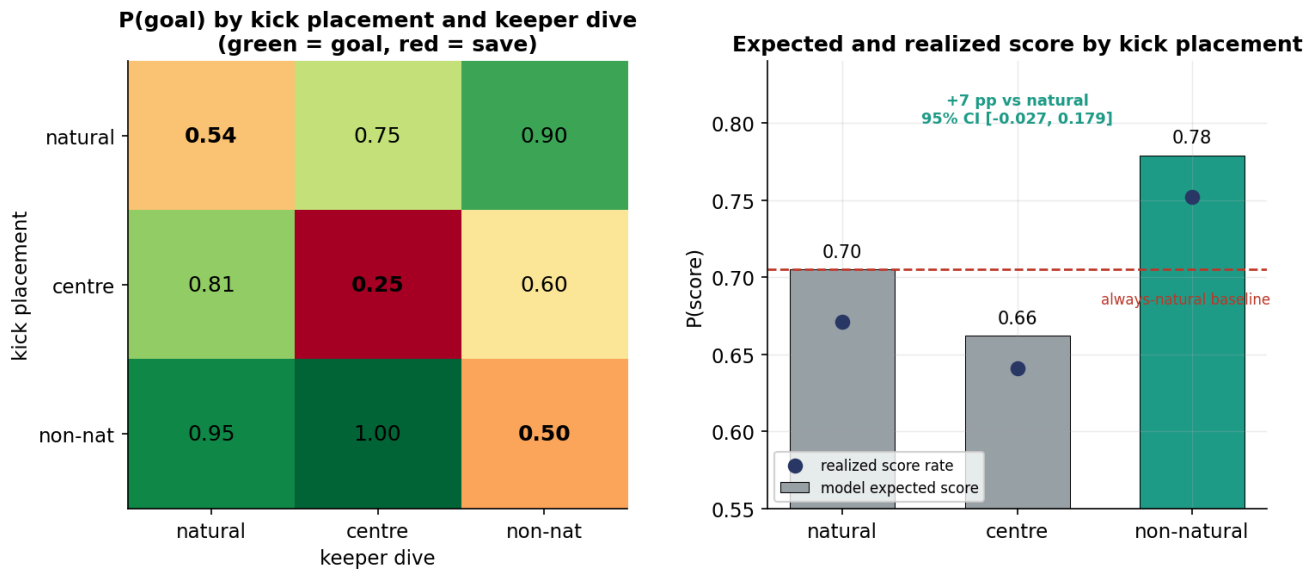


Figure 4: Left: P(goal) by kick placement and keeper dive; the keeper saves mainly on the diagonal. Right: expected score (bars) and realized score (points) by placement, against the always-natural baseline.

This corresponds to roughly eight additional goals per hundred relative to always shooting the natural side, consistent with the long-noted but previously unquantified observation that kickers under-exploit goalkeeper bias. The estimate is limited by sample size: with 35 shootouts the 95% interval on the gain runs from about three goals below zero to eighteen above, so the effect is directionally consistent but not statistically conclusive, and it is a counterfactual that holds goalkeeper behaviour fixed.

4. Why the centre is not the high-value option

A natural objection is that, since keepers commit to a side 88% of the time, the centre should be open. The data do not support this, and the relevant comparison is expected value, which weights each conversion by how often it occurs.

Conditional on the keeper diving away, a central shot scores 0.70 [0.57, 0.80] while a non-natural corner scores 0.96 [0.88, 0.99]: a diving keeper's legs still cover the centre, so vacating a side is not the same as opening the goal. Weighting by frequency, a central kick meets an absent keeper 88% of the time yet converts only 0.70, and meets a stationary keeper 12% of the time, converting 0.25 [0.07, 0.59] (on eight kicks, hence imprecise). The expected score is therefore $0.88 \times 0.70 + 0.12 \times 0.25 \approx 0.66$, below the non-natural corner's 0.78. Even conditional on being on target, the centre converts worst (0.72 against 0.79). The genuinely under-defended region is the non-natural corner, which the keeper reaches on only 46% of such kicks and cannot reach on 96% of his mistakes.

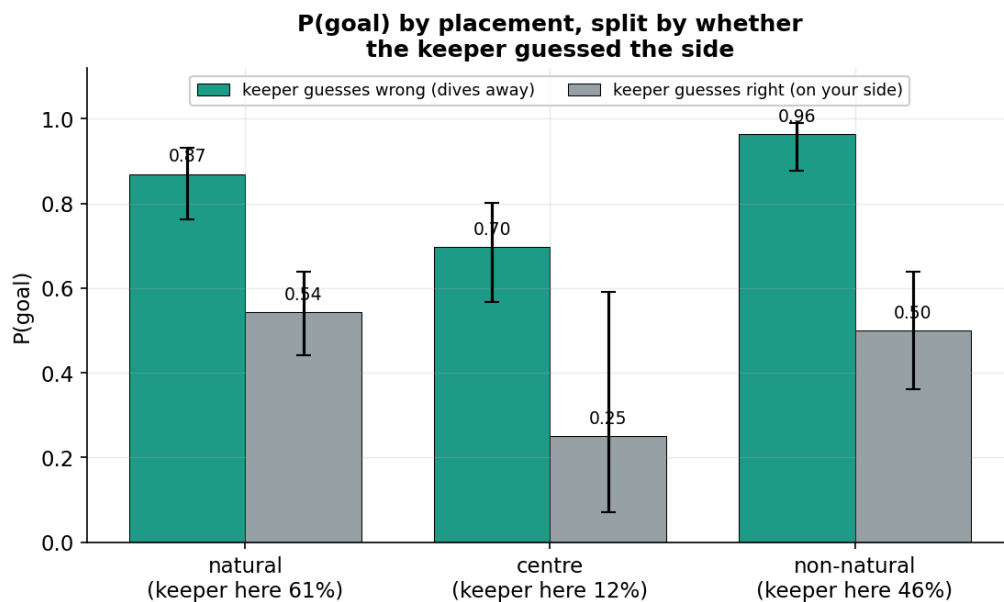


Figure 5: $P(\text{goal})$ by placement, split by whether the keeper guessed the side (teal: guessed wrong; grey: guessed right), with Wilson 95% intervals. The wide centre/right interval reflects only eight kicks.

5. Practical implications

For the penalty taker

Ranked by expected score: non-natural corner (0.78), natural corner (0.71), centre (0.66). The keeper occupies the natural side on 61% of natural-side kicks but only 46% of non-natural kicks, and when he guesses wrong a corner is near-certain whereas the centre is not. Priorities follow: (i) a reliably on-target corner struck with pace is the highest-value skill, since corners convert only when on frame and the non-natural corner is on target 95% of the time; (ii) competence with both corners, because a one-sided taker is readily scouted and the keeper already favours the natural side; (iii) the centre as a low-frequency, disguised option only, given its poor conversion when anticipated. Less skilled strikers should prioritise an on-target corner over the centre, while stronger strikers can place a corner beyond the reach of a correct dive. Against a goalkeeper of exceptional reach the central and mid-height regions shrink fastest, further favouring low, wide placement (a reasoned extension, not measured here).

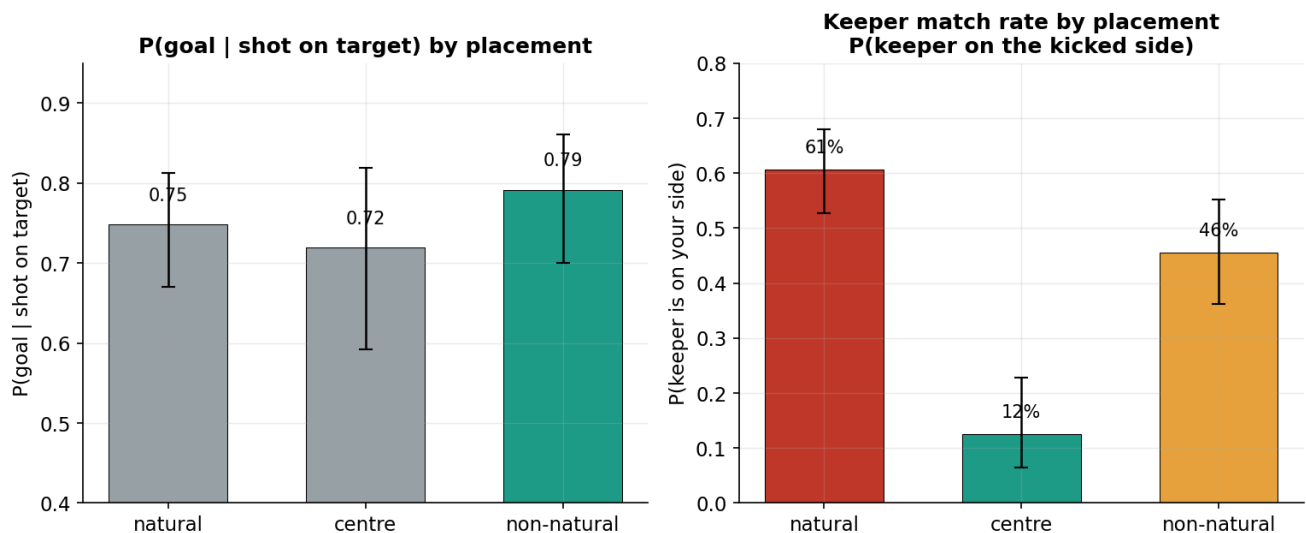


Figure 6: Left: $P(\text{goal} | \text{shot on target})$ by placement; corners convert better than the centre even conditional on hitting the target. Right: the keeper match rate, $P(\text{keeper on the kicked side})$. Each bar is a separate conditional rate, so they need not sum to one.

For the goalkeeper

Within a shootout there is no sequence to exploit, so preparation should target a specific taker's placement distribution rather than the order of kicks. Once committed, guessing the correct side is the sole determinant of a save: a matched dive saves about half of corner kicks, a mismatched dive almost none. Central shots are stopped by the legs even on a committed dive, so the centre should not be abandoned. At the population level goalkeepers may be too predictable: a keeper who always commits, and always to the natural side, is precisely the pattern the kicker analysis exploits, so occasional central

holds or non-obvious choices raise the cost of preparing against him. Training that delays commitment or extends reach converts incorrect guesses into partial saves.

For the analyst and coaching staff

1. **Data.** Placement is available in StatsBomb open data; dive direction is not (the keeper is logged on his line), so video coding, a commercial feed, or a set such as the Kaggle shootouts is required. Code from the goalkeeper's viewpoint and fix the centre-band width in advance.
2. **Conventions.** Reconcile left/right viewpoints before merging sources, and work in natural, centre, and non-natural relative to the foot.
3. **Shrinkage.** Do not use raw per-player frequencies: most takers have fewer than ten penalties on record and raw counts are over-confident. Pool toward the footedness prior and attach a confidence weight; trust a strong individual read only for high-volume takers.
4. **Scope.** Sequential and momentum models are uninformative for both agents here; prioritise placement distributions and reachability.
5. **Application.** For the taker, identify the opposition keeper's directional lean and target the corner he vacates; for the keeper, supply the taker's pooled placement distribution, noting that most takers are near-random and the edge is small.
6. **Uncertainty.** Report intervals and flag thin cells; the "keeper holds the centre" save rate here rests on eight kicks and should be treated as indicative only.

Limitations

The goalkeeper analyses rest on 35 shootouts, so several cells are thin. The headline centre figures (0.64 overall, 0.70 conditional on the keeper diving away) sit on 56 to 64 kicks and are reasonably precise, but the "holds the centre, saves three in four" figure rests on eight kicks and is indicative only. Selection is also present, since takers who choose the centre are not a random sample. The kicker corpus is mildly concentrated (Messi accounts for 9% of penalties), but the placement split is unchanged when high-volume takers are removed, and the Kaggle-based tactical results contain no player identities and survive leave-one-team-out resampling, so neither result is driven by a single player. The exploitation gain is a counterfactual that holds keeper behaviour fixed and its interval still includes zero. Finally, StatsBomb logs no dive direction, so the goalkeeper side cannot be enlarged without additional data.

References

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- Data: StatsBomb Open Data (github.com/statsbomb/open-data); Kaggle, World Cup Penalty Shootouts 1982 to 2022.

Every number is reproducible from the scripts in [scripts/](#).